

SAGE

Synthetic Asset Generation Evidence

Open, vendor-neutral provenance for AI-generated and AI-modified media

White Paper v1.06

Draft architecture and reference specification

1. Executive Summary

SAGE (Synthetic Asset Generation Evidence) is a proposed open, vendor-neutral standard for persistent provenance of AI-generated and AI-modified digital media. Its purpose is not to determine whether media is truthful, fraudulent, harmful, or acceptable. SAGE answers a narrower question: what synthetic-media provenance does this file carry?

The design deliberately minimizes the embedded record. A SAGE payload identifies whether the source was AI-generated before the first participating AI interaction, followed by an ordered chain of AI/provider identifiers and generation identifiers. The first AI record identifies the first participating generator or editor; every later record represents a subsequent AI edit by its position in the chain.

SAGE stores the same core provenance in two places: ordinary file metadata as the fast path, and a redundant concealed, machine-readable image signal as the fallback when metadata has been stripped. Websites can therefore perform pre-submit analysis locally without sending the media to a third party.

2. Goals and Design Principles

- Open and vendor-neutral: no AI company owns the format or decoder.
- Minimal payload: encode only what is necessary to identify provenance.
- Redundant: metadata provides a zero-cost fast path; concealed encoding provides persistence.
- Append-only history: participating AI systems preserve the existing chain and append their own record.
- Local-first detection: receiving sites can inspect media on their own infrastructure.
- Policy-neutral: SAGE reports provenance; the receiving platform decides what to do with it.
- Privacy-conscious: prompts, usernames, account IDs, IP addresses, and other personal data are not part of the public SAGE record.
- Extensible: the logical provenance format is separate from the physical concealment algorithm.

3. Core Provenance Model

The minimum logical payload is:

VERSION | SOURCE_TYPE | AI_ID₁:GEN_ID₁ | AI_ID₂:GEN_ID₂ | ...

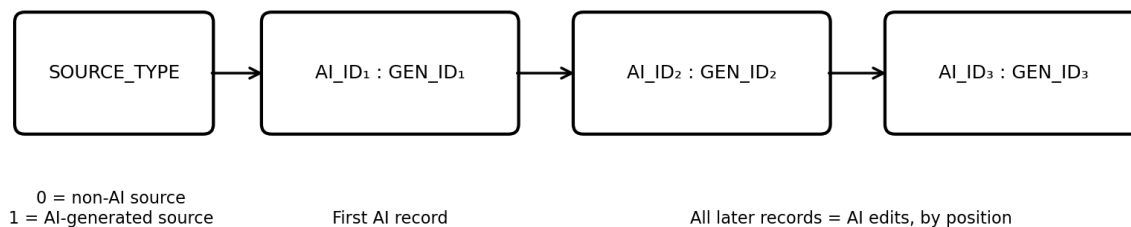
Core/transport boundary: The logical SAGE record contains semantic fields only. Checksums, ECC, redundancy, bootstrap encoding, and physical placement are media-profile transport mechanisms and are not part of the canonical Core serialization unless a future Core extension gives one of them semantic meaning.

SOURCE_TYPE is the one explicit semantic flag required by the chain:

- **SOURCE_TYPE = 0:** the source media was not AI-generated before the first recorded participating AI interaction. The first AI record therefore represents the first AI edit.
- **SOURCE_TYPE = 1:** the source media was AI-generated. The first AI record represents the original AI generation.

No operation flag is required for individual AI records. Chain position provides that meaning: the first record is the first participating AI interaction, and all records after it are later AI edits.

SAGE ordered provenance chain



3.1 AI_ID

AI_ID is a compact identifier for the participating AI system or provider namespace. The SAGE specification should define a collision-resistant registration or naming convention, but the standard itself should remain independent of individual vendors.

3.2 GEN_ID

GEN_ID is an opaque, preferably non-sequential identifier assigned by the participating AI system. It may be resolved by that provider to additional information such as a generation timestamp or internal record, but such resolution is optional and is not required for local SAGE detection.

4. Dual-Layer Encoding

4.1 Metadata layer - fast path

Whenever the file format supports metadata, SAGE writes the logical provenance record into ordinary metadata fields. This costs effectively nothing in image quality and allows extremely inexpensive upload-time detection.

A conceptual metadata representation might be:

```

SAGE-Version: 1
SAGE-Source-Type: 1
SAGE-Chain: OAI:GEN_A | ADB:GEN_B
  
```

4.2 Concealed machine-readable layer - fallback

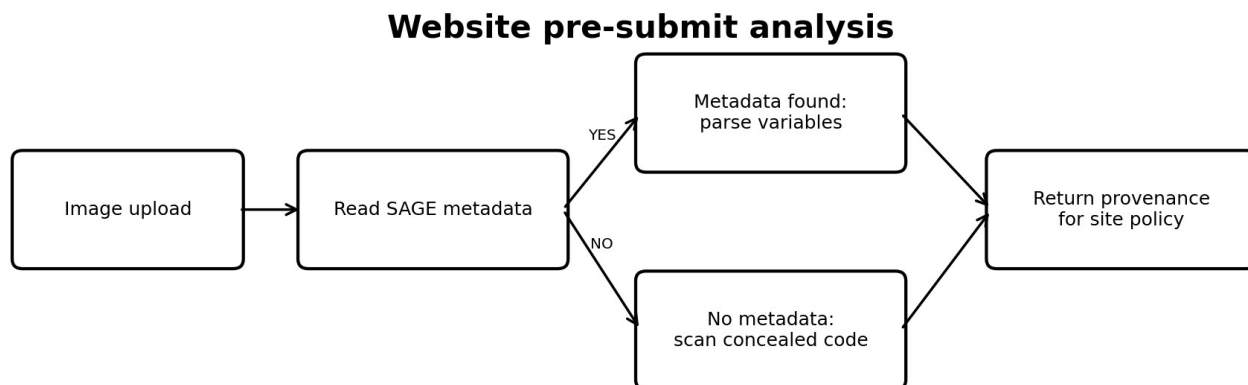
The same core provenance is also embedded into the media using a concealed, QR-like machine-readable signal. The signal should be spatially redundant and use error correction so that enough information may survive normal transformations such as cropping, resizing, recompression, and screenshots.

The concealed representation is not intended to look like a literal QR code. 'QR-like' describes the design philosophy: compact machine-readable symbols, repetition, reconstruction, and error correction. The concealment algorithm is the principal technical research area of SAGE.

4.3 Detection priority

Metadata is the cheapest decoding fast path, but it is never authoritative over the concealed layer. Receiver policy determines whether concealed evidence is also checked. Fast or Normal operation MAY skip a concealed scan for throughput, but any mode that skips it MUST report that concealed-layer integrity was not checked. When both layers are decoded and disagree, the decoder reports CONFLICT rather than silently preferring metadata.

A high-assurance implementation may optionally compare both layers, but this is outside the required fast-path behavior.



5. Generation and Editing Behavior

5.1 AI-generated original

When a participating AI creates new media from synthesis rather than receiving a pre-existing non-AI source, it initializes SOURCE_TYPE = 1 and writes its AI_ID and GEN_ID as record 1.

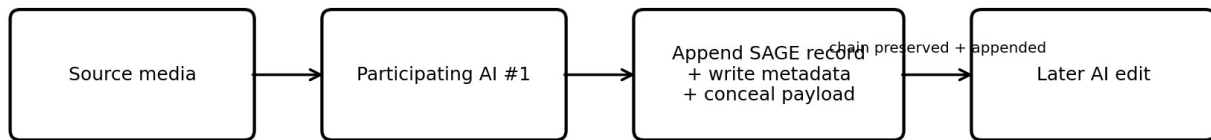
5.2 First AI edit of non-AI media

When a participating AI receives an original image that has not previously been manipulated by a participating AI, it initializes SOURCE_TYPE = 0 and writes its AI_ID and GEN_ID as record 1. This preserves the important distinction between a synthetic original and a real-world source later modified with AI.

5.3 Editing existing SAGE media

A participating AI reads the existing SAGE chain, preserves it in order, appends its own AI_ID:GEN_ID pair, and rewrites both the metadata and concealed representation. Existing valid entries must not be reordered or intentionally removed by a conforming implementation.

SAGE generation and edit flow



6. Website / Platform Integration

SAGE is designed so a website can perform analysis before accepting or publishing an upload. The integration layer can remain extremely small because the decoder reports provenance rather than enforcing policy.

- Read SAGE metadata.
- If SAGE metadata exists and parses successfully, return the provenance variables.
- If SAGE metadata does not exist, run the concealed-signal decoder.
- If a concealed record is recovered, return the provenance variables.
- If neither layer yields a record, return provenance as unknown - never as proof that the media is non-AI.
- Optionally verify AI_ID / GEN_ID pairs with provider services when stronger authentication is desired.

A platform may then label the media, retain provenance, request additional disclosure or verification, flag it for review, or take no action. Those decisions are explicitly outside the SAGE protocol.

7. Reference API Shape

The reference open-source library should expose a small, language-neutral conceptual API:

```
encode(media, source_type, ai_id, generation_id)
append(media, ai_id, generation_id)
read_metadata(media)
scan_concealed(media)
decode(media)
verify(provider_id, generation_id)  # optional
```

Implementations may provide bindings for PHP, Python, JavaScript, C/C++, Rust, Java, or other environments. The concealment engine can be implemented as a native library or command-line utility while higher-level languages use thin wrappers.

8. Concealment Requirements

SAGE v1.0 does not mandate a final concealment algorithm. A candidate algorithm should be evaluated against the following requirements:

- Imperceptibility: ordinary viewers should not see obvious patterns, banding, or artifacts.
- Redundancy: the payload should be distributed across multiple regions rather than stored in one vulnerable location.
- Error correction: partial damage should not automatically destroy the record.
- Crop resistance: a useful fraction of the image should contain enough information for recovery.
- Resize resistance: common downscaling and resampling should preserve recoverable information.
- Compression resistance: common JPEG/WebP recompression should not immediately erase the signal.
- Screenshot tolerance: practical testing should include display-and-recapture workflows.
- Low computational cost: upload-time scanning should be practical on ordinary web infrastructure.
- Open implementation: the encoder and decoder should be auditable and reproducible.

9. Security and Trust Model

SAGE provenance is evidence supplied by participating systems; it is not, by itself, proof that the depicted event or person is real or false. Likewise, absence of SAGE data means only that SAGE provenance was not recovered.

A malicious actor may attempt to forge an AI_ID or GEN_ID. For stronger trust, providers may expose an optional verification mechanism that confirms whether a generation identifier belongs to their namespace. The open SAGE format should not require online verification for basic detection.

Future versions may define cryptographic authentication, signed provider namespaces, or chain-integrity mechanisms. These should be added without forcing large payloads into the concealed channel.

10. Privacy

The public SAGE record should remain deliberately sparse. It should not contain prompts, user identities, account identifiers, IP addresses, private generation parameters, or other personal information. A provider may retain additional internal records under its own policies, but those records are outside SAGE.

GEN_ID values should be opaque and difficult to enumerate. Sequential identifiers should be avoided.

11. Failure and Edge Cases

- Metadata present, concealed signal not checked: normal and expected under the fast-path design.
- Metadata absent, concealed signal recovered: normal recovery behavior.
- Neither present: provenance unknown.
- Malformed metadata: implementation may fall back to concealed scanning.
- Partially recoverable concealed signal: decoder should report measurable recovery diagnostics and a recovery status rather than fabricate missing fields or invent an uncalibrated numeric confidence value.
- Non-participating AI editor: it may preserve the existing signal accidentally or damage/remove it; no new SAGE record is expected.
- Chain growth: v1 implementations should support bounded payload sizes and define a future-compatible strategy if practical limits are reached.

12. Testing Strategy

A clean reference image should be used as the sterile control. An exact copy is passed through the SAGE encoder so that the only intended difference is the provenance embedding.

- Compare sterile and SAGE-encoded images visually and with objective image-difference metrics.
- Verify exact metadata round-trip.
- Verify concealed payload round-trip.
- Test JPEG/WebP recompression at multiple quality levels.
- Test proportional resizing and resampling.

- Test aggressive and partial crops.
- Test screenshots and re-encoding.
- Test sharpening, mild color correction, and common social-media processing.
- Measure false-positive and false-negative rates across a large corpus.

13. Open-Source Reference Project

The reference implementation should be published as an open-source project with a permissive license so AI vendors, social platforms, forums, dating services, marketplaces, CMS developers, researchers, and independent websites can adopt the same format without licensing a proprietary detection service.

A practical repository can be divided into four small components: specification, encoder, decoder, and conformance tests. Provider-specific verification adapters should remain optional.

14. Format-Aware Redundancy

SAGE defines one logical provenance model, but the physical redundancy strategy should adapt to the capabilities of the media format. A static image provides spatial redundancy; a multi-frame asset also provides temporal redundancy. The protocol should use that additional capacity without needlessly duplicating the full provenance record in every frame.

14.1 Static images

For single-frame formats such as JPEG and PNG, the normal SAGE model applies: ordinary metadata is the fast path and a spatially redundant concealed signal is the fallback. Format-specific implementations may use native metadata/chunk facilities where appropriate, while preserving the same logical SAGE fields.

14.2 High-frame and animated media

For animated or high-frame media, SAGE should avoid writing the complete provenance chain into every frame. Lightweight frame marks can carry only the minimum source declaration together with enough SAGE identification to distinguish the signal from coincidental image structure. The complete provenance set is then embedded in selected frames distributed across the duration of the asset.

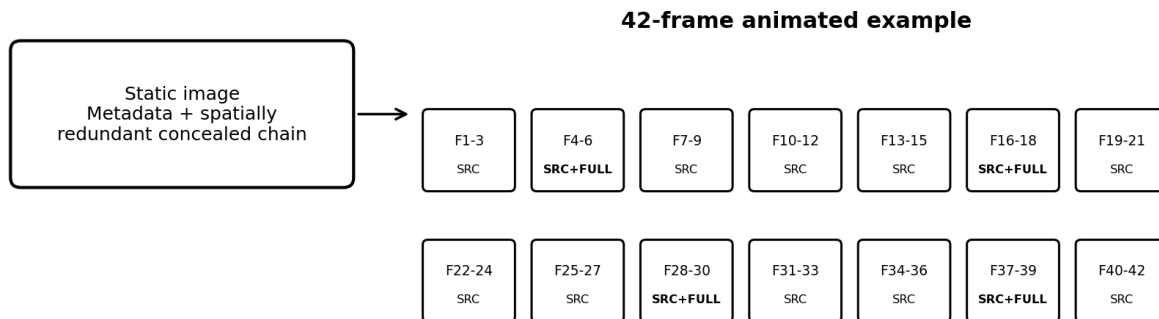
A lightweight frame mark is conceptually: SAGE_MAGIC | VERSION | SOURCE_TYPE. A selected full-record frame is conceptually: SAGE_MAGIC | VERSION | SOURCE_TYPE | AI_ID1:GEN_ID1 | AI_ID2:GEN_ID2 |

Full-record placement should be spread across the timeline rather than clustered. Implementations may use pseudo-random placement derived from stable SAGE data so that distribution is reproducible without storing a separate frame-position table. The density of full-record frames should scale with media length and expected transformation risk.

14.3 Example: 42-frame GIF

A 42-frame GIF can place the lightweight SAGE source marker throughout the animation while placing complete provenance records in only a handful of frames distributed across the 42-frame span. This provides substantial temporal redundancy at very low payload cost. If individual frames are removed or extracted, lightweight provenance may still survive; if the complete animation is available, the decoder can scan until a full record is recovered.

Format-aware SAGE redundancy



Lightweight frames carry SAGE presence/version + SOURCE_TYPE.
Full provenance records are distributed pseudo-randomly across the timeline.

14.4 Redundancy dimensions

- Metadata redundancy: the logical SAGE record is stored in ordinary media metadata when supported.
- Spatial redundancy: concealed information is repeated or distributed across different regions of a frame.
- Temporal redundancy: multi-frame assets distribute lightweight and full records across time.
- Container redundancy: format-native metadata structures may provide additional copies where practical.

The decoder should normally exploit the cheapest available layer first. For a complete file, metadata is the preferred fast path, but receiver policy may require concealed spot-checking or full layer comparison. If metadata is absent, a static asset is scanned spatially; a multi-frame asset may be scanned spatially and temporally until sufficient SAGE data is recovered.

15. Future Media Types

The initial proof of concept can target still images and animated image formats. The logical SAGE chain remains media-independent. Future profiles may define concealment transports for video, audio, animation, and other synthetic assets while preserving the same provenance semantics and format-aware redundancy principles.

16. Non-Goals

- SAGE does not decide whether content is misinformation.
- SAGE does not decide whether an account is fraudulent.
- SAGE does not prove that unmarked media is human-created.
- SAGE does not require publication of prompts or user information.
- SAGE does not require centralized storage of all generations.
- SAGE does not dictate a platform's moderation or labeling policy.

17. Encoder Cost and Decoder Security Profiles

SAGE is designed to be applied at the individual image or frame generation stage. The encoder performs a very small deterministic write while the media is already in memory. Because the logical payload is compact, native generation-time overhead should be extremely small relative to the cost of producing the media itself.

17.1 Encoder behavior

- Write or update the ordinary SAGE metadata fields.
- Write the compact concealed SAGE signal into the generated image or frame.
- For existing SAGE media, preserve the ordered chain and append the current AI_ID:GEN_ID record.
- For high-frame media, use the lightweight per-frame source marker and distribute complete records across the frame span as defined by the applicable format profile.

The encoder does not change behavior according to a receiving site's security preference. It writes the SAGE record once, at generation/edit time, using the applicable media profile.

17.2 Decoder workload

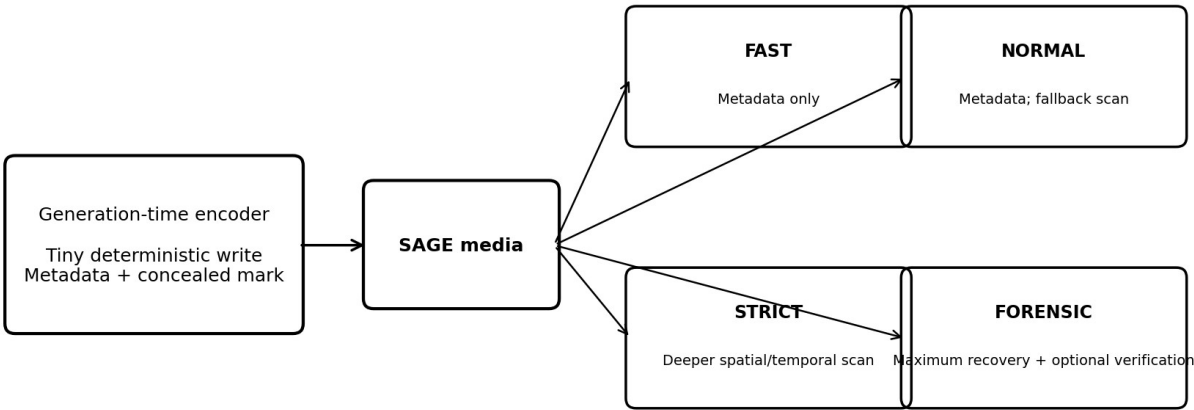
The principal computational tradeoff belongs to the receiving decoder. A website, application, moderation pipeline, or forensic tool may choose how aggressively it searches for and cross-checks recoverable SAGE data. This allows the same media and the same open decoder to serve very different risk environments.

17.3 Suggested decoder security levels

Profile	Behavior	Typical use
Fast	Read metadata first; concealed spot-checking may be disabled or sampled at a very low receiver-defined rate. Metadata-only Fast mode provides no concealed-layer integrity assurance.	Low-risk sites; bulk ingestion where maximum throughput matters.

Profile	Behavior	Typical use
Normal	Read metadata first; if absent or malformed, perform the standard concealed scan. May sample concealed checks even when metadata exists.	General social, forum, CMS, and marketplace uploads.
Strict	Use metadata fast path, then compare concealed provenance whenever practical; perform deeper spatial/temporal recovery when needed.	Identity-sensitive, dating, trust-and-safety, or higher-risk upload contexts.
Forensic	Perform maximum practical recovery and layer comparison, report integrity/confidence, optionally verify provider records, and when warranted seek external corroboration such as provider-side similarity search or retained generation history.	Investigations, research, evidence review, or specialized moderation.

SAGE: lightweight encoding, configurable decoding



Receiving site pays only for the level of assurance it chooses.

17.4 Security profile is a receiver policy

Decoder security levels are implementation and deployment policy, not new provenance semantics. They do not alter SOURCE_TYPE, the ordered AI_ID:GEN_ID chain, or the SAGE encoding written by generators. A receiving site pays only for the degree of recovery and assurance it chooses.

A platform may expose these profiles as configuration presets or may tune the underlying parameters directly, including metadata-only operation, number of spatial regions scanned, number

of animation/video frames inspected, error-correction effort, recovery-quality thresholds derived from measured profile diagnostics, and optional provider verification.

18. Trust Semantics, Threat Model, and Decoder States

SAGE is primarily a standardized declaration and detection mechanism for AI involvement in media. Its provenance chain is evidence about participating tools and transformations, not an absolute or self-authenticating chain of custody. Unless independently verified, an AI_ID:GEN_ID entry is a claim carried by the asset.

18.1 Trust boundary

A conforming decoder **MUST** distinguish detection from authentication. The presence of a syntactically valid SAGE record establishes that SAGE-formatted provenance was detected; it does not, by itself, prove that every claimed provider generated or edited the asset.

Provider verification remains optional. A provider may expose a verification mechanism for its own identifier namespace, and higher-assurance deployments may use it. SAGE Core v1.06 does not require cryptographic signatures or a global certificate authority. Future profiles may add authenticated records without changing the core ordered-chain model.

18.2 Structured decoder observations

The decoder should return structured observations rather than a binary SAGE / no-SAGE answer. Implementations may expose these as fields, flags, or equivalent typed states:

PRESENT: A recoverable, syntactically valid SAGE record was detected.

NOT_DETECTED: No recoverable SAGE signal was found. This **MUST NOT** be interpreted as proof of human origin.

DAMAGED: Fragments or markers indicate SAGE data may be present, but the full record cannot be recovered reliably.

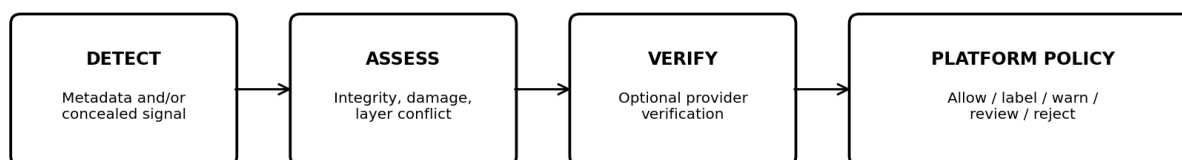
CONFLICT: Two recoverable SAGE layers or records disagree, such as metadata conflicting with the concealed payload.

UNVERIFIED: A provider claim is present but has not been independently verified.

VERIFIED: One or more provider claims were successfully verified through an optional provider mechanism.

These observations are not platform policy decisions. A receiving platform decides whether to allow, label, warn, review, reject, or otherwise handle the asset.

SAGE v1.06: detect, assess, optionally verify, then apply policy



SAGE reports evidence and integrity state. It does not decide whether content is fraudulent, authentic, acceptable, or human-created.

NOT_DETECTED never means HUMAN_CREATED. Provider claims may remain unverified.

18.3 Metadata priority and concealed spot-checking

Metadata remains the cheapest normal decoding path, but it is not inherently authoritative over the concealed layer. Because ordinary metadata can be rewritten easily, deployments MAY perform concealed spot-checks even when valid SAGE metadata is present.

The frequency and depth of spot-checking are receiver policy. Fast deployments may perform no or rare concealed checks; Normal deployments may sample a configurable fraction of assets; Strict deployments should compare layers whenever practical; Forensic deployments should perform maximum practical recovery and comparison. If both layers are decoded and disagree, the decoder should report CONFLICT rather than silently preferring metadata. A Fast deployment with concealed checking disabled knowingly trades concealed-layer integrity assurance for throughput and MUST NOT present metadata-only parsing as concealed-layer verification.

Random or policy-driven spot-checking lets high-volume systems trade compute for assurance without changing the SAGE encoder or payload. The standard does not mandate a universal sampling percentage.

18.4 Deliberate removal and provenance laundering

SAGE distinguishes incidental transformation from adversarial manipulation. Spatial and temporal redundancy are intended primarily to improve survival under cropping, localized edits, frame loss, recompression, resizing, screenshots, and similar handling. A deliberate whole-image purification or reconstruction process may damage every concealed copy simultaneously.

A malicious actor may also strip an existing record and write a new false SAGE record. SAGE v1.06 does not claim to make such provenance laundering impossible. Where a dispute matters, concealed-layer conflicts, provider verification, retained source assets, generation records, and forensic comparison may provide additional evidence. No invisible provenance mechanism should be treated as impossible to remove or forge.

18.5 Composite and multi-source media

Pixel-, object-, or region-level provenance graphs are outside the scope of SAGE Core v1.06. If a non-AI source is subsequently composited with AI-generated or AI-modified material, the resulting asset can still declare AI participation in its ordered SAGE history, but SAGE does not attempt to specify which pixels or regions came from which source. Detailed composite lineage is a possible future extension and is not required for the core objective of detecting AI involvement.

18.6 Transcoding and participating platforms

Cross-format conversion can destroy format-specific metadata and concealed signals. A participating transcoder or platform that detects valid SAGE provenance before conversion SHOULD re-encode the applicable SAGE record into the output format using that format's SAGE profile. A non-participating transcoder may remove SAGE unintentionally; therefore NOT_DETECTED remains an unknown-provenance state, never a declaration of human origin.

18.7 Bootstrap and concealment-version discovery

The concealed transport requires a small stable bootstrap that can be recognized before the decoder knows which concealment algorithm or profile version to invoke. Conceptually, the bootstrap contains a SAGE magic identifier plus a concealment profile/version selector. The detailed bit layout is deferred to the concealment specification, but the bootstrap namespace MUST be designed for coexistence of multiple decoder generations.

A conceptual decode sequence is: detect stable SAGE bootstrap -> identify concealment profile/version -> invoke the appropriate decoder -> recover the logical provenance payload.

18.8 Open algorithm and placement assumptions

SAGE assumes that attackers can know the complete public specification. Frame placement, spatial distribution, error correction, and decoding behavior MUST NOT depend on secrecy of the algorithm. Deterministic or pseudo-random placement may therefore be used for robustness and implementation simplicity, but must not be described as a security secret.

18.9 Verification privacy

Local SAGE detection and decoding should not require network access. Optional provider verification can reveal that a verifier is checking a particular provider identifier or generation record and therefore creates a potential privacy and re-identification channel. Implementations MUST NOT silently treat provider verification as a required phone-home step. Platforms using network verification should disclose and govern that behavior according to their privacy and security requirements.

18.10 Bounded chain growth

The ordered chain can grow as media passes through multiple participating AI systems. SAGE v1.06 recognizes bounded payload capacity as an open engineering constraint. The final byte budget and truncation strategy should be set only after concealment-capacity testing. A future-compatible

strategy should preserve source semantics, explicitly indicate truncation, and avoid silently presenting a partial chain as complete.

19. Forensic Analysis and Lineage Reconstruction

The Forensic profile extends beyond routine SAGE decoding. Its purpose is investigation: when an asset contains damaged, conflicting, incomplete, or apparently falsified SAGE information, an authorized investigator may combine recoverable in-file evidence with external corroborating records. SAGE does not require this external infrastructure and does not treat it as necessary for ordinary detection.

19.1 Asset-centric versus investigation-centric analysis

Strict decoding is primarily asset-centric: it attempts to recover and compare the evidence physically present in the submitted media. Forensic analysis is investigation-centric: it may begin with the same local recovery process and then, when warranted, seek independent evidence outside the asset.

The Forensic profile MAY use provider verification, retained generation or edit records, prior versions, source assets, other SAGE-bearing copies, and provider-side image similarity or reverse-image matching. These sources supplement the embedded record; they do not change the SAGE Core payload.

19.2 Provider-side similarity search

When SAGE evidence indicates AI involvement but the declared AI_ID:GEN_ID chain is missing, damaged, conflicting, or suspected to have been falsified, a participating provider MAY compare the disputed media against generation or editing records it lawfully retains. The provider may use perceptual similarity, reverse-image matching, feature matching, or other appropriate forensic methods.

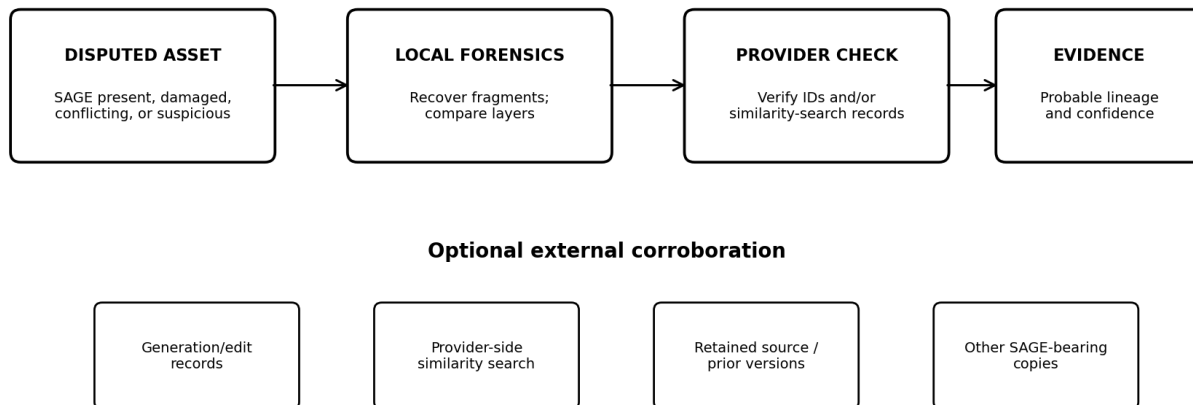
A sufficiently strong match may identify a probable earlier generation or edit record and recover associated provider-side facts such as a generation identifier, processing timestamp, or prior SAGE chain observed at the time of processing. This creates an evidentiary path that is independent of the currently embedded metadata and concealed payload.

19.3 Walking a probable lineage backward

Lineage reconstruction may proceed across providers when records exist. For example, Provider B may retain evidence that its input carried a valid Provider A SAGE entry even if the currently disputed asset no longer contains that entry. Provider B's historical record can therefore help identify an earlier hop, after which Provider A may independently verify or search its own records.

Any reconstructed history MUST be represented as probable or corroborated lineage according to the strength of the available evidence. SAGE does not convert a similarity match into absolute proof of custody.

SAGE v1.06: forensic lineage reconstruction



MATCH_FOUND can add evidence. NO_MATCH proves nothing. External searches are optional, provider-controlled, privacy-sensitive, and limited to lawfully retained records.

19.4 Forensic result semantics

External forensic results should preserve the same observation-versus-verdict separation used by the decoder. A provider or forensic implementation may report results such as MATCH_FOUND, NO_MATCH, RECORD_VERIFIED, PRIOR_CHAIN_FOUND, or INCONCLUSIVE, together with supporting evidence and, where independently calibrated, an evidence-strength estimate. Core decoding does not require a universal numeric confidence score.

MATCH_FOUND can strengthen a lineage hypothesis. NO_MATCH MUST NOT be interpreted as proof that the provider never generated or edited the asset. Records may have expired, been deleted, never been retained, fallen outside the searchable corpus, or become too transformed for reliable matching.

19.5 Retention, participation, and privacy

SAGE does not require providers to retain every generated asset, prompt, user identity, or generation record indefinitely. SAGE neither requires nor encourages providers to retain media or generation records beyond their existing operational, legal, or policy requirements. Forensic similarity analysis operates only over records a provider independently and lawfully retains. Retention remains subject to provider policy, user expectations, applicable law, privacy obligations, and operational constraints. Providers MAY offer forensic search only for records they are permitted and able to retain.

Provider-side similarity search is optional and potentially privacy-sensitive. It should not occur as a hidden part of ordinary local decoding. High-assurance or dispute-driven workflows should make clear when an asset or derived fingerprint is being submitted to an external provider for verification or similarity analysis.

19.6 Generation identifiers as forensic pointers

Opaque generation identifiers remain intentionally compact. Their purpose is not to carry the full forensic record inside the media. When a provider retains supporting records, the generation identifier can act as a provider-scoped pointer to richer evidence. This keeps routine SAGE payloads small while allowing deeper investigation when justified.

19.7 Limits of forensic reconstruction

Forensic reconstruction is evidence accumulation, not guaranteed recovery. A malicious actor may use non-participating tools, reconstruct media, destroy concealed signals, or alter an asset beyond reliable similarity matching. Providers may also lack historical records. SAGE therefore defines forensic reconstruction as an optional escalation path rather than a promise that true lineage can always be recovered.

19.8 White-paper scope and future specification separation

This white paper intentionally presents SAGE Core concepts, media-profile concepts, decoder assurance levels, and optional forensic escalation in one architectural document so the complete model can be reviewed as a system. This does not imply that every future implementer must implement every capability.

When SAGE moves from white-paper architecture to normative implementation specifications, the project SHOULD separate the minimal interoperable core from specialized profiles. A likely document family is: SAGE Core Specification; format-specific image and animated-media profiles; a SAGE Forensic Profile; and a conformance/test suite. An encoder or ordinary receiving platform should be able to adopt the relevant Core and media profile without implementing provider-side forensic infrastructure.

The purpose of this separation is to preserve a small adoption surface while allowing advanced recovery and investigation to evolve independently. Forensic methodology is therefore an optional extension of the SAGE ecosystem, not a prerequisite for basic SAGE conformance.

20. Independent Review Findings and Design Disposition

SAGE has been subjected to independent architectural critique during the white-paper stage. The purpose of this section is not to reproduce individual reviews, but to record the major recurring findings, the project's disposition of each finding, and the reasoning behind decisions to adopt, defer, or explicitly exclude proposed changes. This creates a design record so that future implementers can distinguish deliberate tradeoffs from overlooked problems.

Review findings are classified as: ADOPTED when they exposed a weakness that should change the architecture or its stated guarantees; SCOPED when the issue is real but is intentionally outside the guarantees of SAGE Core; DEFERRED when a decision depends on experimental capacity or implementation data; and PROFILE / EXTENSION when the capability is useful but should not enlarge the minimal interoperable core.

20.1 Detection is not authentication

Finding: Unsigned AI_ID:GEN_ID records can be forged, and the mere presence of a SAGE-formatted record cannot prove that a named provider generated or edited the asset.

Disposition: ADOPTED as a trust-semantic correction. SAGE explicitly distinguishes detection from authentication. PRESENT means SAGE-formatted evidence was detected; provider claims may remain UNVERIFIED. Optional provider verification can raise assurance. Mandatory signatures, PKI, certificate rotation, revocation infrastructure, and global trust stores are not required by Core because SAGE's primary objective is interoperable evidence of AI involvement, not cryptographically guaranteed chain of custody. Authenticated provider records remain a possible future extension.

20.2 Metadata downgrade and layer conflict

Finding: Easily rewritten metadata can falsely override a truthful concealed signal if a decoder treats metadata as authoritative and skips concealed analysis.

Disposition: ADOPTED. Metadata is a computational fast path, not an authority. Receivers may perform concealed spot-checking even when metadata is valid, and a decoded disagreement between metadata and concealed provenance is reported as CONFLICT. Sampling frequency remains receiver policy so high-volume platforms can choose their own assurance/throughput balance. Metadata-only Fast mode explicitly provides no concealed-layer integrity assurance.

20.3 Cross-asset replay and media binding

Finding: A valid SAGE payload could be copied from one asset and re-encoded into an unrelated asset. Metadata and concealed layers could agree, and a provider could confirm that the referenced GEN_ID exists even though it belongs to a different image.

Disposition: OPEN / DEFERRED for Core 0.1 research. This is a genuine semantic attack distinct from ordinary metadata forgery. A conventional cryptographic image hash is poorly suited to SAGE because benign transformations such as recompression, resizing, screenshots, and minor pixel changes would invalidate exact hashes. Perceptual fingerprints are promising but introduce their own interoperability and adversarial-threshold problems. Candidate solutions include provider-side asset-aware verification, profile-level perceptual binding, or a standardized verification result such as ASSET_MATCH / ASSET_MISMATCH / INCONCLUSIVE. SAGE v1.06 does not prematurely mandate a media hash.

20.4 Linear history versus composites and local edits

Finding: Real media workflows can form directed graphs rather than linear histories, including inpainting, masked edits, blending, and composites made from several AI and non-AI sources.

Disposition: SCOPED. SAGE Core answers the simpler question of whether AI participated in the resulting asset and preserves an ordered participating history where possible. Pixel-, region-, mask-, object-, and DAG-level provenance are intentionally non-goals for v1.06. A detailed composite-

lineage extension may be defined later without requiring every ordinary image to carry a provenance graph.

20.5 Chain growth and payload capacity

Finding: Append-only provenance can grow beyond the physical capacity of a robust concealed channel.

Disposition: DEFERRED pending empirical capacity testing. SAGE recognizes that the chain must ultimately be bounded, but setting an arbitrary entry count or byte budget before the concealment experiments would optimize against imaginary constraints. A future truncation rule should preserve source semantics, explicitly mark truncation, and never silently present a partial history as complete.

20.6 Concealed-signal feasibility

Finding: The concealed channel is load-bearing but not yet demonstrated. Simultaneously achieving imperceptibility, capacity, crop resistance, resizing resistance, recompression resistance, screenshot resistance, and practical decoding cost is a significant watermarking/steganographic research problem.

Disposition: ACCEPTED AS THE PRIMARY ENGINEERING RISK. The white paper does not claim this problem is solved. The next experimental phase must establish real recovery rates and capacity before a normative media profile is locked. If acceptable robustness cannot be demonstrated, the media-profile architecture must change rather than hiding the failure behind stronger prose.

20.7 Cropping, canvas expansion, and coordinate changes

Finding: Extreme crops, borders, aspect-ratio changes, outpainting, rotations, and canvas expansion can defeat concealment schemes tied to original dimensions, normalized grids, or global transforms.

Disposition: DEFERRED TO MEDIA-PROFILE EXPERIMENTS. The reference tests should deliberately include severe crops, padding, outpainting, rotation, resizing, and aspect-ratio changes. Concealment must not rely on secrecy or on preservation of the original coordinate frame.

20.8 Temporal trimming and animated media

Finding: If full provenance records appear only in selected frames, a short clip or trimmed animation can retain lightweight AI markers while losing every frame containing the full chain.

Disposition: ADOPTED AS AN ANIMATED-PROFILE DESIGN REQUIREMENT, with the exact mechanism deferred. Candidate approaches include distributed reconstruction fragments, erasure coding, small chain identifiers, and other temporal redundancy schemes that permit lineage recovery from useful subsets of surviving frames. Detection of AI involvement and recovery of full lineage remain distinct outcomes.

20.9 Transcoding and non-participating pipelines

Finding: CDNs, messaging systems, social platforms, and editors routinely strip metadata, resize media, and transcode formats; non-participating pipelines may destroy both SAGE layers.

Disposition: ADOPTED AS AN INTEROPERABILITY LIMIT. Participating transcoders should detect SAGE before conversion and re-encode the applicable record into the output profile. SAGE cannot require non-participating software to preserve it. Consequently, NOT_DETECTED always means provenance unknown/not recovered and never proves human origin.

20.10 Decoder cost and high-volume deployment

Finding: Deep concealed scanning across very large upload volumes may be computationally expensive and may push platforms toward metadata-only operation.

Disposition: ADOPTED AS AN EXPLICIT TRADEOFF rather than treated as a protocol defect. Receiver-controlled Fast, Normal, Strict, and Forensic profiles let platforms choose assurance appropriate to workload and risk. The encoder remains independent of receiver policy. Real decoder benchmarks will be required before recommended profile defaults are standardized.

20.11 Provider verification privacy and availability

Finding: Online verification can leak upload/browsing activity, add latency, create availability dependencies, and fail permanently if a provider shuts down or deletes records.

Disposition: ADOPTED. Local SAGE detection requires no network access. Provider verification is optional and privacy-sensitive, must not silently phone home, and should return observation-level evidence rather than a platform verdict. Provider disappearance can leave historical claims permanently UNVERIFIED; SAGE does not pretend otherwise. Future authenticated/offline extensions may reduce this dependency but are not Core requirements.

20.12 GEN_ID enumeration and verification abuse

Finding: Predictable generation identifiers or unrestricted verification endpoints could permit enumeration, scraping, replay, or denial-of-service attacks.

Disposition: PARTLY ALREADY CONSTRAINED / IMPLEMENTATION SECURITY. Generation identifiers should be opaque and non-sequential. Verification services, when offered, require ordinary abuse controls such as rate limiting, authorization where appropriate, non-enumerating responses, and privacy-preserving API design. These service controls do not need to enlarge the embedded SAGE payload.

20.13 Deliberate purification and provenance laundering

Finding: An attacker may globally reconstruct or purify an asset to remove concealed signals, or strip a legitimate chain and write a false replacement.

Disposition: SCOPED AS AN EXPLICIT LIMITATION. Redundancy is intended to survive common transformations and partial damage, not to make watermark removal mathematically impossible. Conflict detection, provider verification, retained evidence, similarity search, and forensic reconstruction may expose manipulation, but SAGE does not promise that true lineage can always be recovered.

20.14 Forensic similarity search and lineage reconstruction

Finding: When embedded records are falsified or damaged, providers may possess independent historical evidence capable of reconstructing probable lineage.

Disposition: ADOPTED AS AN OPTIONAL FORENSIC PROFILE. Providers may use verification, lawfully retained generation/edit records, prior SAGE-bearing copies, and perceptual/reverse-image similarity analysis. SAGE creates no new retention requirement. MATCH_FOUND can add evidence; NO_MATCH proves nothing. Forensics is investigation-centric and remains outside the minimal adoption surface.

20.15 Bootstrap and decoder-version discovery

Finding: A decoder must recognize enough of SAGE to discover which concealment algorithm/profile version to invoke, creating a bootstrap problem if version information exists only inside the payload that requires that decoder.

Disposition: ADOPTED AS A PRE-NORMATIVE REQUIREMENT. The concealed transport needs a small stable bootstrap namespace containing a recognizable SAGE identifier and concealment profile/version selector. Its exact bit layout is intentionally not frozen until the media experiments provide capacity and robustness data. This bootstrap is a long-lived compatibility boundary and should receive focused review before Core/media-profile lock.

20.16 Why the white paper remains broader than Core

Finding: Adding forensic methodology and specialized recovery behavior risks making a supposedly lightweight standard increasingly large and difficult to adopt.

Disposition: ADOPTED AS DOCUMENT ARCHITECTURE. The white paper intentionally shows the complete SAGE ecosystem, but future normative specifications will be separated structurally. SAGE Core will contain only transport-independent semantics; media-specific concealment belongs in media profiles; forensic investigation belongs in a Forensic Profile; and interoperability should be tested through a conformance suite. Basic adopters should not need to implement the entire ecosystem.

20.17 Review convergence and next-stage decision

Independent reviews converged on several recurring pressure points: authentication limits, metadata spoofing, composite workflows, bounded chain capacity, transcoding, verification privacy, and concealed-signal robustness. This convergence is treated as evidence that these are genuine design pressures rather than reviewer-specific preferences. The current architecture addresses them by correcting trust semantics, exposing limitations, or assigning unresolved questions to the layer where empirical evidence is required.

The most important unresolved conceptual issue entering Core Specification 0.1 is cross-asset replay/media binding. The most important unresolved engineering issue is the concealed media profile itself. The white paper should therefore remain stable while these questions move into

normative Core design and controlled image/animation experiments rather than accumulating speculative mechanisms.

21. v1.06 Proposed Conformance Summary

Requirement	v1.06 behavior
Source declaration	SOURCE_TYPE: non-AI source or AI-generated source
History	Ordered AI_ID:GEN_ID chain
Operation semantics	Implicit from chain position; no per-entry operation flag
Metadata	Write/read SAGE fields when format supports them
Concealed encoding	Redundant QR-like machine-readable fallback
Decoder priority	Metadata first; concealed scan only when metadata is unavailable or malformed
Platform policy	Outside the protocol
Provider verification	Optional
Personal data	Excluded from the public payload
Implementation	Open, vendor-neutral reference code
Format-aware redundancy	Static: spatial fallback; multi-frame: lightweight source marks plus distributed full records
Generation-time cost	Compact deterministic metadata + concealed write at individual image/frame generation
Decoder assurance	Receiver-configurable; suggested Fast, Normal, Strict, and Forensic profiles
Trust semantics	Detected provenance is evidence/claim unless independently verified; detection is not authentication
Decoder states	Expose PRESENT, NOT_DETECTED, DAMAGED, CONFLICT, UNVERIFIED, and VERIFIED or equivalent structured observations
Layer conflict	If metadata and concealed provenance are both decoded and disagree, report conflict; do not silently prefer metadata
Concealed spot-checking	Receiver-configurable even when valid metadata exists
Bootstrap/versioning	Stable concealed bootstrap identifies SAGE and selects concealment profile/version
Adversarial removal	Explicitly outside absolute guarantees; redundancy improves recovery but cannot guarantee survival against deliberate purification
Composite media	Region/pixel-level provenance graphs are a v1.06 non-goal

Requirement	v1.06 behavior
Transcoding	Participating transcoders should re-encode detected SAGE provenance into the output profile
Verification privacy	Local decode requires no network; provider verification is optional and privacy-sensitive
Fast profile assurance	Metadata-only Fast mode is an explicit throughput tradeoff and provides no concealed-layer integrity assurance
Forensic escalation	May combine local SAGE recovery with optional external corroboration
Provider similarity search	Optional provider-side reverse-image/perceptual matching against lawfully retained generation/edit records
Forensic no-match semantics	NO_MATCH is inconclusive and never proves absence of provider involvement
Lineage reconstruction	Recovered external evidence may establish probable/corroborated prior hops without claiming absolute custody
Provider retention	SAGE does not require indefinite retention of assets, prompts, identities, or generation records
No new retention requirement	SAGE neither requires nor encourages retention beyond a provider's independently established lawful policy
Specification modularity	Future normative specifications should separate the minimal Core/media profiles from optional forensic methodology
Cross-asset replay	Recognized as an open Core 0.1 issue; no premature exact/perceptual hash requirement
Animated trimming	Future animated profile must distinguish persistent AI detection from recoverable full lineage and test temporal subset recovery
Concealment feasibility	Treated as the primary empirical engineering risk; normative media profile waits for measured robustness/capacity
Review disposition record	Major independent-review findings are documented with adopted, scoped, deferred, or extension rationale

22. Implementation Architecture and Reference Algorithm Plan

This section defines the implementation-document architecture that follows the SAGE v1.06 white-paper freeze. It is intentionally pre-normative: it describes how SAGE should move from architectural semantics into reproducible encoder and decoder behavior without prematurely fixing concealment details that still require experiments.

22.1 Two independent reference algorithms

The initial implementation work is divided into two language-neutral reference algorithms: `sage_encode_v0.01.alg` and `sage_decode_v0.01.alg`. They are versioned independently because encoding and decoding have different responsibilities and may mature at different rates. A stable encoder format may remain unchanged while later decoder revisions improve recovery from crops, recompression, resizing, screenshots, temporal loss, or damaged redundancy.

22.2 Purpose of the .alg documents

The .alg documents are not source-code files tied to PHP, Python, C, JavaScript, or any other implementation language. They are precise, language-neutral algorithm specifications expressed as structured pseudocode. Their purpose is to let independent implementations reproduce the same logical behavior and to make algorithm changes reviewable independently of any reference codebase.

Each .alg document should identify at minimum: algorithm name; algorithm version; experimental/stable status; applicable SAGE Core version; applicable media profile; inputs; outputs; required preconditions; deterministic processing steps; error/observation behavior; compatibility expectations; and associated conformance vectors.

22.3 `sage_encode_v0.01.alg` responsibilities

The encoder is deliberately narrow and deterministic. It accepts media plus the SAGE facts supplied by the participating generator/editor, preserves any recoverable conforming prior history, constructs the logical record defined by SAGE Core, writes the ordinary metadata representation, encodes the concealed representation according to the applicable media profile, applies required redundancy/error correction, and emits the resulting media.

The encoder must not redefine SAGE semantics. Definitions such as `SOURCE_TYPE`, `AI_ID`, `GEN_ID`, ordered-chain meaning, truncation markers, and extension semantics belong to SAGE Core. Likewise, pixel placement, redundancy, bootstrap encoding, and format-specific transport belong to the applicable media profile. `sage_encode_v0.01.alg` describes the deterministic procedure that applies those definitions.

22.4 `sage_decode_v0.01.alg` responsibilities

The decoder is expected to carry substantially more recovery complexity than the encoder. It inspects ordinary metadata, searches for the stable SAGE bootstrap where required by the selected assurance profile, identifies the concealment profile/version, recovers spatial or temporal fragments, applies error correction/reconstruction, parses the logical record, compares independently recovered layers, and returns structured SAGE observations. A failed bootstrap inspection is not equivalent to a successful inspection that finds no bootstrap: registry or inspection failure must remain distinguishable and must not collapse to ordinary `ABSENT` or `NOT_DETECTED`. For v0.01, unresolved concealed inspection is reported as `DAMAGED` with an explicit machine-readable inspection-failure diagnostic.

The decoder must not convert evidence into a platform verdict. PRESENT, NOT_DETECTED, DAMAGED, CONFLICT, VERIFIED, UNVERIFIED, recovery quality, and measurable recovery diagnostics are interpreted according to SAGE Core and the applicable media profile. Moderation outcomes such as allow, label, warn, review, or reject remain receiver policy. Optional provider verification and forensic escalation are invoked only where the receiver profile and privacy policy permit them.

22.5 Independent algorithm versioning and backward recovery

Encoder and decoder versions are intentionally decoupled. A later sage_decode_v0.01.alg revision should, where practical, continue decoding media produced by earlier compatible encoder/profile versions. Decoder improvements that recover an existing signal more reliably do not require changing the encoder or the embedded payload format. This permits recovery research to advance without forcing previously encoded assets to become obsolete.

Conversely, an encoder revision that changes the physical transport or bootstrap must identify a compatible media-profile/concealment version so decoders can select the correct recovery procedure. Algorithm version numbers therefore describe implementation behavior; they do not silently redefine SAGE Core semantics.

22.6 Proposed algorithm header

A reference .alg file should begin with a small machine- and human-readable declaration similar to the following conceptual form:

```
SAGE-ALGORITHM
Name: sage_encode
Version: 0.01
Status: Experimental
Profile: SAGE-IMG-PNG-EXP
Core-Version: 0.01
```

```
INPUT:
    media
    source_type
    ai_id
    generation_id
```

```
OUTPUT:
    sage_media
```

```
PROCEDURE:
    ...
```

22.7 Separation of semantics, transport, and procedure

Implementation work should maintain a strict boundary between three layers. SAGE Core defines what fields, states, and ordered history mean. Media Profiles define how those semantics are transported through a particular class of media, including metadata mapping, bootstrap, concealment, redundancy, error correction, spatial/temporal behavior, and capacity limits. The .alg documents define the procedural steps used to encode or recover those profile-defined representations.

A practical design test is whether a rule would remain true if SAGE were transported through a completely different medium. If so, it likely belongs in Core. If it depends on pixels, frames, JPEG/PNG structures, spatial transforms, temporal fragments, or a specific error-correction mechanism, it belongs in a media profile or its encoder/decoder procedure.

22.8 Reference implementations

Reference source-code implementations may be developed in one or more languages, but source code is subordinate to the Core, profile, .alg documents, and conformance vectors. A language implementation must not become the de facto specification merely because it was written first. Independent implementations should be able to reach interoperable results without copying internal behavior from the reference implementation.

22.9 Conformance vectors and sterile-source corpus

The implementation phase should maintain deterministic test vectors. A known sterile source asset, known SAGE input values, and a defined encoder/profile version form the baseline. Where the profile is deterministic, independent conforming encoders should produce equivalent logical and physical SAGE representations. The corresponding decoder must return the defined observation set.

The test corpus should then contain transformed descendants of the encoded baseline, including recompression, resize/resampling, crop, screenshot, WebP/AVIF/JPEG transcoding, blur, sharpening, color changes, borders/padding, rotation, aspect-ratio changes, outpainting/canvas expansion, partial occlusion, and - for animated media - frame deletion and temporal clipping. Each test asset should have an expected decoder result and measured recovery outcome.

22.10 Experimental measurements

The first media-profile experiments should measure rather than assume the central engineering tradeoffs: payload capacity; visible distortion; bootstrap recovery; full-chain recovery; error-correction overhead; crop survival; recompression survival; resize survival; screenshot survival; coordinate-change survival; decoder CPU/GPU cost; and false-positive/false-negative behavior. Metrics such as PSNR and SSIM may supplement visual inspection, but successful SAGE recovery under transformation is the primary protocol measurement.

22.11 Cross-asset replay as a parallel Core question

Cross-asset replay/media binding should be investigated in parallel with concealment rather than solved inside the first pixel algorithm. The initial encoder and decoder should preserve an extension path for later asset-relation verification, but v0.01 should not embed an arbitrary exact or perceptual hash merely to close the issue superficially. Candidate Core/Profile work includes asset-aware provider verification and structured outcomes such as ASSET_MATCH, ASSET_MISMATCH, and INCONCLUSIVE.

22.12 Initial implementation deliverables

The recommended first implementation set is: SAGE Core Specification Draft 0.01; an experimental still-image media profile beginning with PNG; sage_encode_v0.01.alg; sage_decode_v0.01.alg; a

minimal reference encoder/decoder implementation; a sterile-source and transformed test corpus; and a machine-readable conformance-results file. Animated/high-frame media should follow after still-image experiments establish realistic capacity and recovery behavior.

22.13 Promotion criteria

The v0.xx algorithm series is experimental. Promotion toward v1.00 should require a frozen compatible Core/profile version, reproducible independent implementations, published test vectors, measured transformation robustness, documented capacity limits, known failure modes, and a conformance suite capable of detecting incompatible implementations. Version 1.00 should indicate demonstrated interoperability, not merely that the pseudocode appears complete.

22.14 Immediate engineering sequence

The immediate sequence is therefore: freeze the v1.06 architectural white paper; draft Core 0.01 semantics; define the minimum experimental PNG transport/bootstrap needed to run tests; write `sage_encode_v0.01.alg` and `sage_decode_v0.01.alg` against that profile; implement both; generate deterministic baselines; apply the transformation corpus; measure recovery and visual impact; and revise the experimental media profile/algorithms from evidence. Architectural changes to the white paper should occur only if implementation evidence exposes a genuine assumption that cannot be resolved at the Core/profile/algorithm layer.

22.15 Canonical evidence-state vocabulary and resolution

Before independent implementations are expected to conform, layer-level evidence uses one canonical four-state vocabulary: ABSENT, VALID, DAMAGED, and INVALID. ABSENT means no SAGE evidence was detected in that layer/profile. VALID means one complete syntactically valid logical SAGE record was recovered. DAMAGED means evidence indicates SAGE is or was likely present but no complete valid record can be recovered. INVALID means data explicitly claims or structurally resembles SAGE for the selected profile but violates the applicable syntax/profile rules. NONE and NOT_FOUND are not alternate layer states in v0.01; an empty inspection returns an explicit ABSENT evidence object rather than null.

When several applicable profiles or layers are inspected, the decoder collects every complete VALID logical record and deduplicates records by canonical logical equality. Zero distinct valid records are resolved using DAMAGED, INVALID, and ABSENT evidence. One distinct valid record yields PRESENT. Two or more distinct valid records yield CONFLICT. No profile-priority rule, metadata preference, recovery score, or “best” selector may silently suppress a different complete valid record.

22.16 Canonical two-layer evidence matrix

Metadata	Concealed	Same record?	Final presence	Interpretation
ABSENT	ABSENT	n/a	NOT_DETECTED	Nothing recovered; never implies human origin.
VALID	ABSENT	n/a	PRESENT	Metadata-only evidence; concealed integrity not established.
ABSENT	VALID	n/a	PRESENT	Concealed-only evidence.
VALID	VALID	Yes	PRESENT	Layers agree on the same logical record.
VALID	VALID	No	CONFLICT	Preserve both candidate records.
DAMAGED	ABSENT	n/a	DAMAGED	Incomplete SAGE evidence.
ABSENT	DAMAGED	n/a	DAMAGED	Incomplete SAGE evidence.
INVALID	ABSENT	n/a	DAMAGED	Claimed/SAGE-like data violates profile syntax.
ABSENT	INVALID	n/a	DAMAGED	Claimed/SAGE-like data violates profile syntax.
VALID	DAMAGED	n/a	PRESENT	Valid record plus damaged secondary evidence.

Metadata	Concealed	Same record?	Final presence	Interpretation
DAMAGED	VALID	n/a	PRESENT	Valid record plus damaged secondary evidence.
VALID	INVALID	n/a	PRESENT	Valid record plus invalid secondary evidence.
INVALID	VALID	n/a	PRESENT	Valid record plus invalid secondary evidence.
DAMAGED	DAMAGED	n/a	DAMAGED	No complete valid record.
INVALID	INVALID	n/a	DAMAGED	No complete valid record.
DAMAGED	INVALID	n/a	DAMAGED	No complete valid record.
INVALID	DAMAGED	n/a	DAMAGED	No complete valid record.

22.17 Recovery diagnostics instead of uncalibrated confidence

The v0.01 decoder does not require a universal confidence value from 0.0 to 1.0. Until empirical transformation testing provides a calibration basis, such a number would imply mathematical precision the implementation has not earned. Profiles instead expose measurable recovery diagnostics where available, such as bootstrap hits, regions scanned, fragments found and required, redundant copies found and validated, corrected symbol count, checksum result, and decode runtime. A coarse recovery-quality value MAY be reported as EXACT, CORRECTED, PARTIAL, or not applicable. Future specifications may define calibrated confidence semantics if experiments demonstrate that they are useful and reproducible.

22.18 Canonical decoder result object

Every conforming decoder call should return the same result-object skeleton, using null, empty arrays, or empty objects for non-applicable values rather than omitting fields unpredictably. The initial mandatory fields are: algorithm; algorithm_version; mode; status; error_code; error_details; presence; detected; record; candidate_records; integrity; metadata_evidence; concealed_evidence; verification; provider_verification; asset_relation; profile_ids; recovery_quality; recovery_metrics; warnings; errors; semantic_notes; and forensic. This stable envelope allows independent

implementations and test harnesses to compare behavior without depending on language-specific exception or null conventions.

23. Conclusion

SAGE proposes a deliberately small standard: identify the source class, preserve an ordered chain of participating AI systems and generation identifiers, store it cheaply in metadata, and redundantly conceal the same provenance inside the media. Encoding occurs at individual image/frame generation with extremely small native overhead; the receiving decoder controls how much work to spend recovering and cross-checking that provenance according to its own security posture.

The central engineering challenge for the proof of concept remains the concealment layer: finding a machine-readable encoding that is visually unobtrusive yet resilient enough to survive common transformations. SAGE v1.06 keeps its trust boundary explicit: detection is not authentication, NOT_DETECTED is not proof of human origin, and deliberate removal or false provenance cannot be made impossible. Independent review has identified cross-asset replay/media binding as an unresolved Core-design question and temporal trimming, coordinate changes, capacity, and decoder cost as concrete media-profile test requirements. The implementation plan now separates transport-independent semantics from media transport and from the independently versioned `sage_encode_v0.01.alg` and `sage_decode_v0.01.alg` reference procedures. The next stage is therefore Core Specification Draft 0.01 plus an experimental PNG profile, deterministic encoder/decoder algorithms, and a measured transformation corpus rather than further speculative expansion of the architecture.